LOT Pager ��� 03-MECHANICAL-BODY

03 · Mechanical Design — Stainless Body

1. Target

Silver flat square **40 × 40 mm**, nominal **5 mm** height, two stainless-steel parts: - **Part A — Back (mirror):** polished 316L plate, #8 mirror finish, outside face unmarked except laser-etched serial on inside. - **Part B — Front (bezel/tray):** machined 316L tray holding PCB, battery; openings for screen, camera lens, button. Joined by 4 micro-screws (M1.0, hidden under bezel lip) or laser-weld (production, DVT+).

2. Thickness stack — the honest numbers

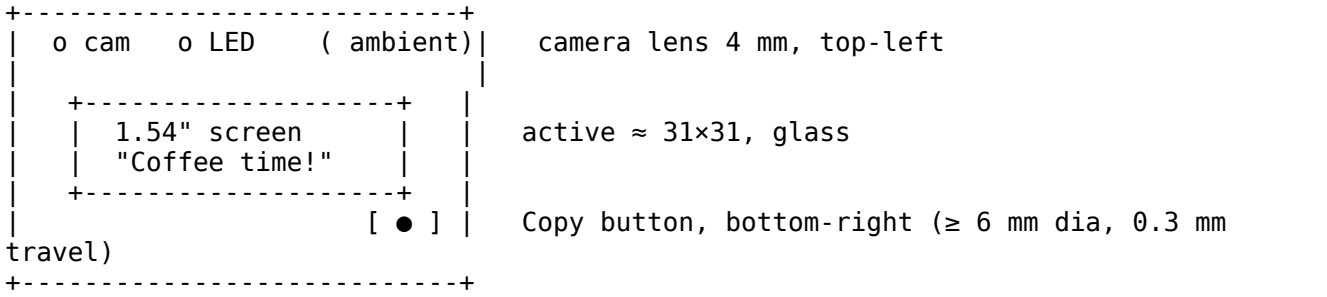
Layer	mm (typ.)
Back plate	0.6
Battery (LiPo 3.0)	3.0
PCB (4-layer)	0.8
Screen + cover glass (front)	1.6 + 0.5
Front bezel wall	0.5
Clearances	0.3
Stack, battery under PCB	**≈ 7.3–7.5 mm**

- Tallest items: camera OV2640 module (≈ 4 mm), ESP32-S3-WROOM (3.2 mm), battery.
- **EVT target: 7.5 mm.** Not 5.
- **DVT path to ~6 mm:** bare-die ESP32-S3 (0.9 mm), camera CSP on rigid-flex beside (not under) the battery, 2.5 mm cell, 0.4 mm back plate, 0.3 mm glass, side-by-side layout (battery next to PCB).
- **5 mm is achievable only without camera** (screen + 2.5 mm cell + thin PCB) — recommend offering as “LOT Pager Slim” variant.
- Decision for Vadik: keep camera at ~6–7.5 mm, or ship 5 mm without camera?

3. RF and wireless-charging windows (critical, R2)

Stainless steel is a conductor: Wi-Fi/BLE antenna and Qi coil inside a closed steel box will not work. - **Antenna:** keep the ESP32 antenna at a PCB edge facing a **1.0–1.5 mm dielectric seam** (PEEK/PC gasket) running along one side of the case between Part A and Part B. Alternative: ceramic chip antenna + front glass window area. Verify with VNA/range test. - **Qi coil:** place ⌀ 25–28 mm coil behind a **ceramic or sapphire disc** inset in the back mirror (circular, flush-polished, looks like a deliberate jewel). Cost ≈ $2–6. Mirror remains “polished stainless” around it. - **Fallback if Qi fails:** two gold pogo pads on the seam for a magnetic dock — still “wireless” to the user (no cable), but not Qi. - Test plan (P3): aluminium shell with 3 window variants; measure RX power at 5 mm coil spacing, RSSI at 3 m.

4. Front face layout (40 × 40)



Button: flush metal dome over tact switch **or** side-mounted tact; waterproofing via silicone membrane (IP54 goal).

5. Materials & finish

- **316L** (corrosion-resistant, skin-safe, polishes well). 304 acceptable for EVT.
- Back: grind → polish → #8 mirror; optional PVD anti-fingerprint.
- Front: bead-blast satin vs back mirror (contrast), or mirror both (brand decision, see brand.lot-systems.com).
- PCBWay CNC supports stainless + polishing; request DFM review for 0.5 mm walls (thin; consider 0.6 min).

6. Drawings to produce (HW-02/03)

STEP model (Fusion/FreeCAD), 2D drawings with tolerances ±0.05, window insert cut-outs, gasket profile, assembly exploded view.